

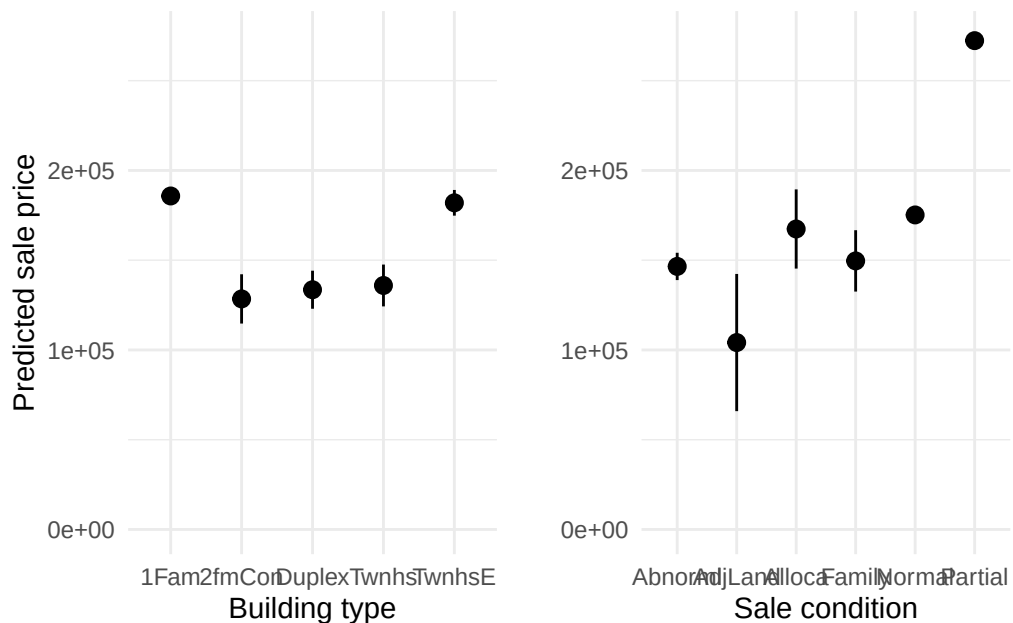
Modeling AMES Housing Data

DATA 608 Spring 2025 - Story 5

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Using the AMES housing dataset, we will be examining how various factors can affect home prices and using them to make predictions for future sales.

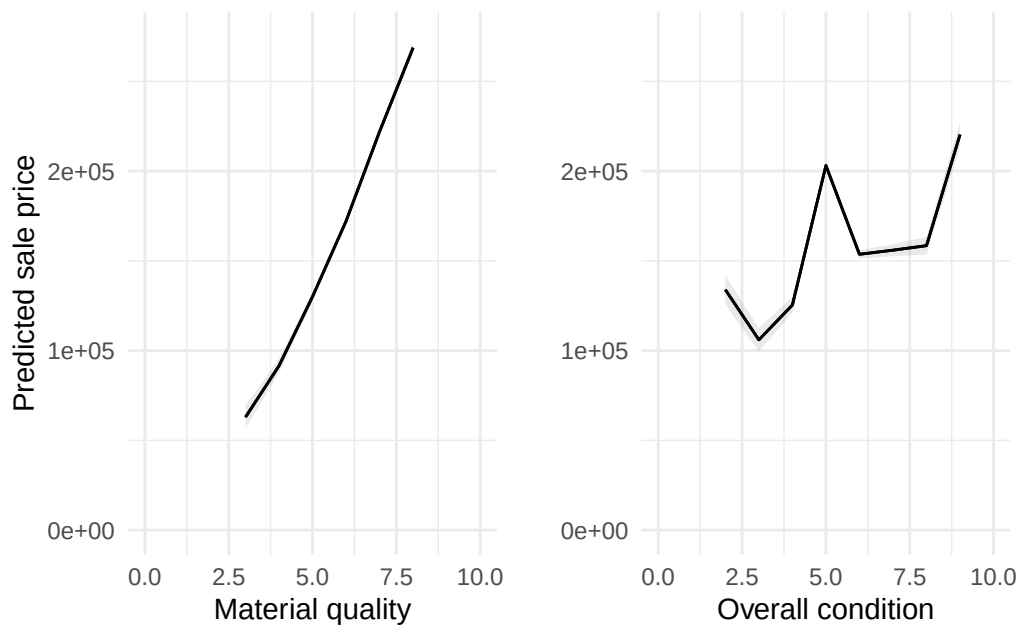
We can begin with a couple of fairly obvious factors that contribute to higher sale prices: building type and sale condition.



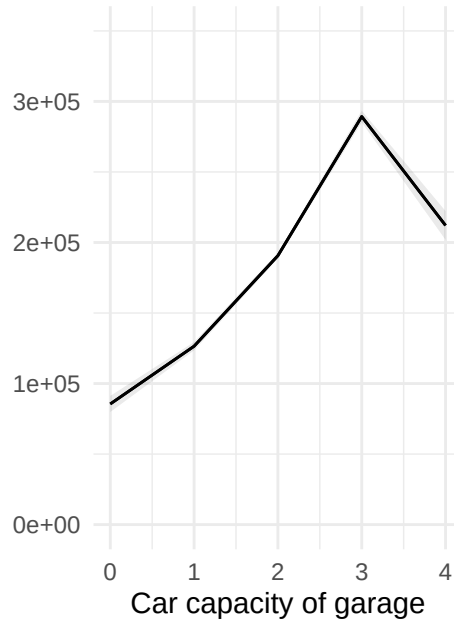
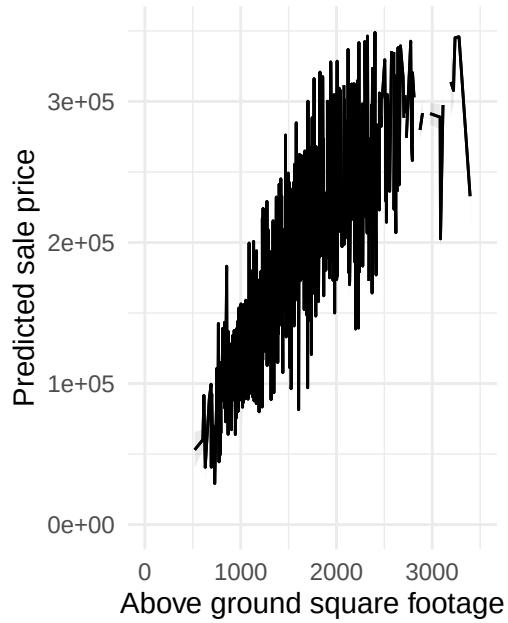
Single-family detached homes and end-units in townhouse rows can be expected to sell for much higher than other types of buildings.

And we can predict sales of “partial” incomplete homes (referring to brand new properties) to be nearly \$100K more than the next category.

High ratings for the quality of the materials and finish of the home can increase the sales price more predictably than higher ratings for the overall general condition of the house.



As expected, square footage of living area increases the predicted price. However, the number of cars that can be accommodated by the garage peaks as a predictor at 3 (this may be due to limited data).



Finally, a less obvious feature is central air, which appears to be associated with a significant boost of almost \$100k. The effect of the quality of the home's heating, in contrast, is less remarkable.

